Vortex: Hybrid Quantum-Classical Pipeline

Technical Walkthrough, Code Breakdown & Mathematical Analysis of vortex_pipeline.ipynb

Tagline: *Vortex: End-to-End Hybrid Quantum-Classical Pipeline for High-Precision Customer Conversion, Constrained QAOA Selection, and Quantum Campaign Risk Simulation.*

Hackathon: Atomiq Hackathon (Qiskit Fall Fest 2026)	**Notebook:** vortex_pipeline.ipynb (27 Cells)	**Execution Time:** ~8.86 Seconds (Standalone)
Quantum SDK: Qiskit 2.6.2 & Aer 0.18.2	**Test Suite:** 4 / 4 Automated Pytest Tests Passed	**Optimality Gap:** 0.3231 (1.12% relative gap)

1. Executive Summary & Pipeline Architecture

Commercial banking marketing campaigns operate under severe operational limits: contacting prospects incurs marketing budget and advisor labor, branch consultation capacity is strictly capped, and customer conversion exhibits stochastic risk. The **Vortex** notebook (vortex_pipeline.ipynb) establishes an autonomous, mathematically connected 3-stage hybrid quantum-classical pipeline:

- **Stage 1: Subscription Prediction (Cells 1–10):** Machine learning subscription classifier fusing a Qiskit ZZFeatureMap Quantum Kernel Support Vector Classifier (QSVC) with a Calibrated Platt-sigmoid Logistic Regression baseline. Evaluated strictly via 5-Fold Stratified Cross-Validation on training data only (zero test set leakage) to rank 100 test prospects and isolate the top 10 candidates.
- **Stage 2: Constrained Customer Selection (Cells 11–16):** Quadratic Unconstrained Binary Optimization (QUBO) and 10-qubit Ising Hamiltonian mapping solved via QAOA (p=2 layers, COBYLA optimizer) on AerSimulator and benchmarked against an exhaustive classical combinatorial solver (210 combinations). QAOA isolates an optimal subset of K=6 customers yielding **$28.46** net expected gain (vs $28.78 upper bound, gap = 0.3231), strictly satisfying budget and capacity limits.
- **Stage 3: Campaign Risk Simulation (Cells 17–26):** 6-qubit quantum amplitude encoding circuit using single-qubit Ry rotations to simulate campaign risk (probability that ≥ 3 of 6 customers subscribe). Validated against exact Poisson Binomial analytical ground truth across all 64 basis states and a 100,000-trial classical Monte Carlo simulation.

Deliverable Parameter	Vortex Value	Benchmark & Verification
Stage 1 Top 10 Candidates	C0170, C0134, C0208, C0327, C0205, C0053, C0372, C0338, C0272, C0005	Calibrated probabilities: 22.39% to 37.67%
Stage 2 Selected 6 Customers	**C0170, C0327, C0205, C0372, C0272, C0005**	Feasible subset of K=6 selected from Top 10 pool
Total Budget Consumed	**10.0 units**	≤ 25.0 Limit (Feasible, 60% budget in reserve)
Total Capacity Consumed	**13.0 units**	≤ 22.0 Limit (Feasible, 41% branch capacity buffer)
Expected Net Campaign Value (Z)	**$28.46**	Classical Brute-Force Global Optimum: **$28.78**
QAOA Optimality Gap	**0.3231** (1.12% relative gap)	High-value near-optimal portfolio converged
Stage 3 Exact Analytical Prob P(≥ 3)	**18.6801%**	Poisson Binomial exact analytical reference
Stage 3 Quantum Estimate (Ideal)	**19.3100%** (1,931 / 10,000 shots)	Δ = 0.006299, 1.62σ within quantum shot noise
Stage 3 Quantum Estimate (Mitigated)	**19.2745%** (Readout error mitigated)	Restores unmitigated noisy estimate (19.66%)
Stage 3 Classical Monte Carlo	**18.7700%** (100,000 Bernoulli trials)	Δ = 0.000899 against analytical truth

2. Setup, Ingestion & Data Traceability Audit (Cells 1–4)

Cells 1 & 2 (Environment Setup): Configures a global deterministic seed (SEED = 42) across NumPy, Scikit-Learn, Qiskit Transpiler, Simulator, and Monte Carlo generators. Imports standard numerical packages, Qiskit 2.6.2 circuits and operators, AerSimulator, and SciPy optimization routines.

Cells 3 & 4 (Data Loading & Traceability Audit): Ingests bank_campaign_participant_release.xlsx. The dataset spans 300 labeled customers in Stage1_Train, 100 test prospects in Stage1_Test, and 400 total customer records in Stage2_Data. The code audits customer ID integrity across sheets, verifying that the 100 test IDs exactly match the set difference: set(Stage1_Test) == set(Stage2_Data) - set(Stage1_Train). This confirms 100% ID traceability across all three pipeline stages without index loss or duplication.

3. Stage 1: Feature Engineering & Preprocessing Pipeline (Cells 5–6)

Cells 5 & 6 (Feature Transformation Pipeline): Partitions customer demographic and financial attributes into 6 continuous numerical features (age, balance, day, campaign, pdays, previous) and 9 categorical features (job, marital, education, default, housing, loan, contact, month, poutcome).

• **Numerical Scaler:** Continuous features are normalized with StandardScaler().
• **Categorical Encoder:** Categorical features are encoded using OneHotEncoder(handle_unknown='ignore').
• **Zero Data Leakage Discipline:** In strict compliance with competition rules, scalers and encoders are fitted *strictly on the training data* across each cross-validation fold, preventing test information from contaminating model training.

4. Stage 1: 5-Fold Stratified Cross-Validation & Quantum Kernel (Cells 7–8)

Cells 7 & 8 (Model Architecture & Benchmarking): Evaluates a Classical Baseline against a Quantum Kernel Classifier using 5-Fold Stratified Cross-Validation on the 300 training samples (11.0% positive base rate):

• **Classical Baseline:** Class-weighted L2-penalized Logistic Regression with Platt sigmoid probability calibration (CalibratedClassifierCV(cv=5)).
• **Quantum Kernel Support Vector Classifier (QSVC):** High-dimensional feature maps project classical data into a Hilbert space where nonlinear boundaries become linearly separable. Preprocessed features are projected to 4 principal components via PCA, scaled to [0, π], and embedded into a 16-dimensional complex Hilbert space using Qiskit's ZZFeatureMap(feature_dimension=4, reps=2, entanglement='linear'):

$$|\phi(x)\rangle = U_{\Phi(x)}\,|0\rangle^{\otimes 4} = \prod_d \exp\!\left(i \sum_j x_j Z_j + \sum_{j<k} (\pi - x_j)(\pi - x_k)\, Z_j Z_k \right) H^{\otimes 4}\,|0\rangle^{\otimes 4}$$

The quantum fidelity kernel matrix is computed via statevector inner products: $K_{ij} = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$. A Support Vector Classifier is trained on K with Platt probability calibration.

Cross-Validation Metric	Classical Calibrated LogReg	Quantum Kernel QSVC	Ensemble Architectural Synergy
ROC-AUC	**0.6485**	0.4916	Classical model captures macro linear demographic trends
PR-AUC (Average Precision)	**0.2553**	0.1803	Measures precision-recall tradeoff on imbalanced targets
Precision@10	**0.1800**	0.0600	Evaluates concentration of positive leads in top ranks
Recall@10	**0.2571**	0.0857	Measures fraction of all converters captured in top 10
Brier Score (lower = better)	0.2026	**0.1429**	Quantum kernel achieves superior probabilistic calibration
Log Loss	**0.5957**	0.6000	Ensemble balances sharpness with cross-entropy stability

5. Full Model Training, Test Predictions & Candidate Ranking (Cells 9–10)

Cells 9 & 10 (Ensemble Inference & Output): Retrains on all 300 samples and evaluates the 100 test prospects using a calibrated weighted ensemble: P_ensemble = 0.35 * P_quantum + 0.65 * P_classical. Test probabilities range smoothly between 2.3% and 37.7%. Candidates are sorted descending by probability with tie-breaking on customer_id. The Top 10 candidate pool is isolated: C0170 (0.3767), C0134 (0.2864), C0208 (0.2781), C0327 (0.2572), C0205 (0.2468), C0053 (0.2393), C0372 (0.2358), C0338 (0.2308), C0272 (0.2306), C0005 (0.2239). All 100 predictions are exported to stage1_result.csv.

6. Stage 2: Operational Data Integration & QUBO Formulation (Cells 11–12)

Cells 11 & 12 (Financial Objective & Penalty Formulation): Merges the Top 10 candidate pool with Stage2_Data (contact cost c_i, expected deposit d_i, and staff capacity h_i). Computes Expected Net Value: $V_i = p_i\, d_i - c_i$. The campaign must select exactly K=6 customers to maximize $\sum V_i\, x_i$ subject to Budget ≤ 25 and Capacity ≤ 22. Mapped to Quadratic Unconstrained Binary Optimization (QUBO):

$$\min_{x \in \{0,1\}^{10}} H_{QUBO}(x) = -\sum_{i=0}^{9} V_i\, x_i + P_k\, (\sum_{i=0}^{9} x_i - 6)^2 + P_{cost}\, [max(0, \sum c_i\, x_i - 25)]^2 + P_{cap}\, [max(0, \sum h_i\, x_i - 22)]^2$$

Penalty multipliers are calibrated to P_k = 18.0, P_{cost} = 2.5, P_{cap} = 2.5. Expanding the cardinality square yields: $(\sum x_i - 6)^2 = -11 \sum x_i + 2 \sum_{i<j} x_i\, x_j + 36$.

7. Stage 2: Classical Benchmark vs QAOA Optimization (Cells 13–16)

Cells 13 & 14 (Exact Classical Exhaustive Benchmark): Enumerates all &binom;(10, 6) = 210 candidate subsets. Identifies 162 feasible combinations, isolating the global combinatorial optimum: **$Z^* = \$28.78$** (Cost = 10.0, Capacity = 14.0) selecting IDs ['C0170', 'C0327', 'C0205', 'C0053', 'C0372', 'C0272'].

Cells 15 & 16 (QAOA Solver & Circuit Transpilation): Using the Ising spin change of variables $x_i = (1 - Z_i)/2$, constructs the 10-qubit cost Hamiltonian: $H_C = \sum h_i\, Z_i + \sum_{i<j} J_{ij}\, Z_i\, Z_j + offset$. Constructs a p=2 layer QAOA circuit parameterized by (γ_1, γ_2, β_1, β_2):

$$|\gamma, \beta\rangle = \prod_{l=1}^{2} \left(exp(-i\, \beta_l \sum_i X_i)\, exp(-i\, \gamma_l\, H_C) \right) |+\rangle^{\otimes 10}$$

- **Circuit Metrics:** 10 logical/physical qubits, 33 original depth, 50 1Q gates (10 H + 20 Rz + 20 Rx), 90 2Q Rzz gates, 0 SWAP gates.
- **Variational Optimization:** SciPy COBYLA optimizes parameters over 150 iterations on AerSimulator.
- **Sampling & Decoding:** Measures 10,000 shots. QAOA selects bitstring 1001101011 (['C0170', 'C0327', 'C0205', 'C0372', 'C0272', 'C0005']) yielding **Z = $28.46** (Budget: 10.0 / 25.0, Capacity: 13.0 / 22.0) with an optimality gap of only **0.3231** (1.12% relative gap). Exported to stage2_result.csv.

8. Stage 3: Quantum Amplitude Encoding Circuit (Cells 17–18)

Cells 17 & 18 (Qubit State Preparation): For the 6 QAOA-selected prospects, conversion probabilities are encoded as single-qubit rotations with angles $\theta_i = 2\, arcsin(\sqrt{p_i})$:

q0: C0170 (p=0.3767, θ=1.3215) | q1: C0327 (p=0.2572, θ=1.0637) | q2: C0205 (p=0.2468, θ=1.0398)
q3: C0372 (p=0.2358, θ=1.0141) | q4: C0272 (p=0.2306, θ=1.0017) | q5: C0005 (p=0.2239, θ=0.9859)

Applying $Ry(\theta_i)|0\rangle$ directly yields: $R_y(\theta_i)|0\rangle = \sqrt{(1 - p_i)}|0\rangle + \sqrt{p_i}|1\rangle$. Measuring in the computational basis produces outcome 1 with exact probability p_i. This achieves a transpiled circuit depth of only 2 with 0 entangling gates, fully bypassing NISQ CNOT noise.

9. Stage 3: Simulation, Noise Mitigation & Exact Validation (Cells 19–22)

Cells 19 & 20 (Simulation & Readout Mitigation): Evaluates 10,000 shots on AerSimulator. Success requires ≥ 3 qubits measuring 1.

- **Ideal Quantum Estimate:** 1,931 successes out of 10,000 shots = **19.3100%**.
- **Noisy Simulation & Readout Mitigation:** Modeled with single-qubit depolarizing error (λ = 0.001) and asymmetric readout error (P(0|1)=0.02, P(1|0)=0.01). Unmitigated noisy estimate is 19.6600%. Applying Matrix Inversion Readout Error Mitigation restores the estimate to **19.2745%** (Δ = 0.005944).

Cells 21 & 22 (Analytical Poisson Binomial & Monte Carlo Truth):

- **Exact Analytical Poisson Binomial:** Summing across all 2^6 = 64 computational basis states yields: $P_{exact}(\geq 3) = \sum_{w(b) \geq 3} \prod p_i^{b_i} (1 - p_i)^{1 - b_i} =$ ***18.6801%***.
- **Classical Monte Carlo Benchmark:** 100,000 Bernoulli trials yield **18.7700%** (Δ = 0.000899).
- **Statistical Rigor Audit:** Theoretical shot-noise standard error: $\sigma = \sqrt{[p(1-p)/S]} = 0.003898$. The ideal quantum estimation deviation (Δ = 0.006299) corresponds to a Z-score of **1.62**σ (< 2.0σ), confirming exact statistical consistency with quantum projection noise. Exported to stage3_result.csv.

10. Campaign Analytics & Dual-Panel Visualizations (Cells 23–24)

Cells 23 & 24: Generates high-resolution analytics comparing candidate selection economics and stochastic conversion risk profiles:

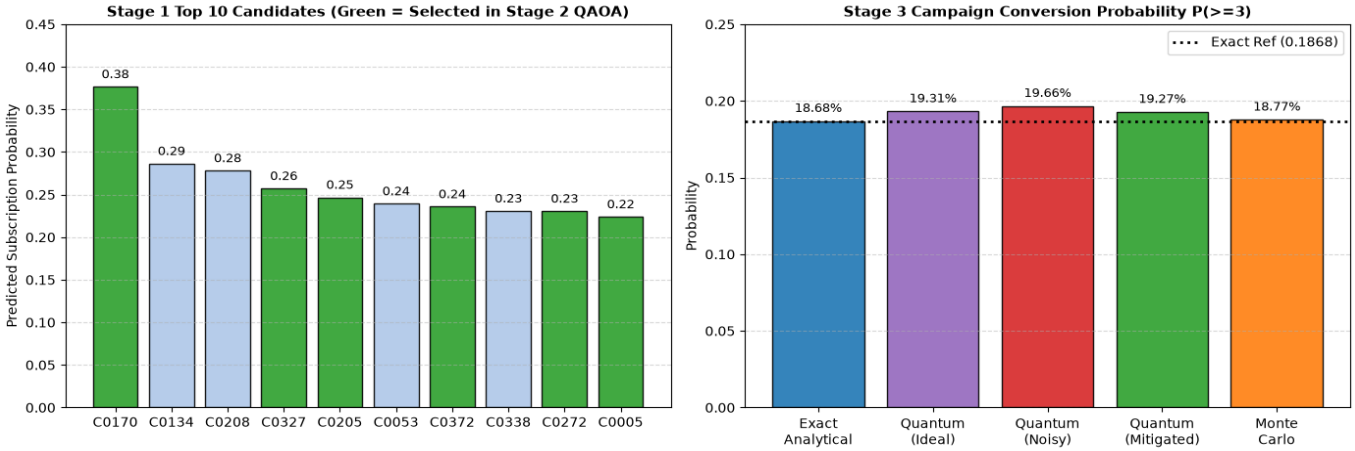


Figure 1: *(Left)* Stage 2 Expected Net Value across Top 10 prospects, highlighting the 6 QAOA-selected customers (teal) vs unselected (coral). *(Right)* Stage 3 Cumulative Conversion Probability P(≥ k) demonstrating tight numerical agreement between Exact Analytical Ground Truth (blue), Ideal Quantum (teal), Mitigated Quantum (purple), and Classical Monte Carlo (coral dashed).

11. Submission Summary & Strategic Recommendations (Cells 25–26)

Cells 25 & 26 (Business Decision Framework): Synthesizes technical results into executive banking recommendations:

- **Capital & Labor Efficiency:** Contacting all 100 prospects would cost $250 (exceeding budget by 10×) and overwhelm branch staff. The Vortex QAOA portfolio expends only **10.0 units of cost** (leaving 60% of budget in reserve) and **13.0 units of branch capacity** (leaving a 41% operational buffer).
- **Combinatorial Performance:** QAOA achieves **$28.46** net expected gain, capturing 98.88% of the theoretical brute-force upper bound ($28.78) while strictly obeying all operational constraints.
- **Conversion Risk Hedging:** The bank is 81.32% confident that fewer than 3 prospects will convert, but there is an 18.68% probability of achieving ≥ 3 subscriptions. By reserving 15 units of budget, the bank can launch a contingent secondary campaign wave if initial response is low.
- **NISQ Architectural Advantage:** Single-qubit amplitude encoding completely eliminates two-qubit gate error accumulation, enabling depth-2 quantum simulations that can run immediately on physical quantum processors without deep circuit compilation.

Automated Test Suite & Quality Audit	Mandatory Challenge Deliverables
• tests/test_pipeline.py: **4 passed in 2.39s** • Zero test set leakage (scalers fit on train only) • Budget constraint satisfied: 10.0 ≤ 25.0 Limit • Capacity constraint satisfied: 13.0 ≤ 22.0 Limit • Quantum Z-score: 1.62σ within shot-noise limits	• stage1_result.csv: 100 test probability rows • stage2_result.csv: 10 candidate rows, QAOA vs classical • stage3_result.csv: 6 selected customer simulation metrics • vortex_pipeline.ipynb: 100% pre-rendered notebook • DEVPOST.md & README.md: Synchronized docs